

Type: VSC1 (Vocational and Skill Enhancement Courses)	
B.Sc(ECS)-II (Semester III)	
Course Title: Based on DSC1-3 and DSC1-4	(Paper Code:)
Total Lectures: 60 Hrs.	Contact Hrs. (P) : 4
University Evaluation: 30 Marks	Internal Evaluation: 20 Marks
Array:	
1.	Write a program that finds only even elements in an array.
2.	Write a program that finds only odd elements in an array.
3.	Write a program that finds the addition of matrices.
4.	Write a program that finds multiplication of matrices.
5.	Implementation of searching and sorting algorithms.
Stack:	
1.	Write a program to check entered string is palindrome or not by using stack.
2.	Write a program to convert decimal number into binary number by using stack.
3.	Write a program to count total number of vowels present in string by using stack.
4.	Write a program which convert infix expression into prefix expression.
5.	Write a program which convert infix expression into Postfix expression.
6.	Write a program which check entered expression is valid or not.
7.	Write a program for evaluation of postfix expression.
8.	Write a program to calculate the factorial of the entered number by using recursion.
9.	Write a program to calculate the digit sum of the entered number by using recursion.
10.	Write a program to find face value of entered number by using recursion.
Queue:	
1.	Write a program to implement Priority queue.
2.	Write a program to implement IRD (Input Restricted Deque)
3.	Write a program to implement ORD (Output Restricted Deque)
Linked List:	
1.	Write a program to implement queue by using linked list. (Dynamic implementation)
2.	Write a program to implement doubly linear linked list with its basic operations.
3.	Write a program to implement singly circular linked list with its basic operations.
4.	Write a program to implement doubly circular linked list with its basic operations.
Tree:	
1.	Write a program to implement BST with the following operations: I) Insert Node II) Count Leaf nodes

	III) Count Non-Leaf nodes IV) Count Total nodes
2.	Write a program to implement BST with the following operations: I) Insert Node III) Find Minimum Node V) Display only odd nodes VII) Display leaf nodes IX) Find the degree of node II) Find Maximum node IV) Search node VI) Display only even nodes VIII) Find the level of node X) Delete Node

Subquery and Join:

A) Table Structure of Account_Master

Field Name	Datatype	Size	Description
Accent	Varchar2	6	Primary Key
Acc_name	Varchar2	6	Account Holders name
Gender	Char	1	Either M or F
Op_Date	Date		The date on which the Account is opened
Chq_fac	Char	1	Cheque facility statue set Default 'N'
Cl_Date	Date		

B) Table Structure of Account_Transaction

Field Name	Datatype	Size	Description
Accent	Varchar2	6	Foreign Key
Tr_amount	Number	7,2	Transaction Amount
Tr_date	Date	1	Transaction Date
Chq_no	Varchar2	10	Cheque Number
Tr_type	Char	1	Receipt or Withdrawal

1.	Display all the details of account holders in ascending order of Op_date.
2.	Display the details of account holders who are having cheque facility and balance >4000.
3.	Display the details of account holders whose balance is between 4000 and 9000.
4.	Display the name and number of female account holder.
5.	Display the details of account holders whose 2nd char of name 'a'.
6.	Display the account of those who have deposited.
7.	Display the details of account holders those who have opened account in 2012.
8.	Display the no of female account holders having cheque facility.
9.	Display the details of 2nd min balance of account holder.

10.	Display the account_no, account_nm, transaction amount, transaction_type of all account holders.
11.	Display the details of account holders who is having maximum balance.
12.	Display the details of account holders whose month of opening the account and transaction are same.
13.	Display the details of account holders who have not made any transaction and whose name is starts with 'R'.
Synonym, Index, View:	
1.	Create a synonym for table student.
2.	Apply DML command on synonym.
3.	Destroy synonym.
4.	Create a view read only view.
5.	Create updateable view.
6.	Apply DML commands on updatable view.
7.	Create read only view for single table.
8.	Destroy view.
9.	Create a sequence for roll_no field in student table.
10.	Insert sequence into table.
11.	Create a cyclic sequence.
12.	Destroy sequence.
13.	Create a cluster table.
14.	Apply DML command on cluster table.
15.	Destroy cluster table.
16.	Create a simple index.
17.	Display an index.
18.	Create composite index.
19.	Create reverse index.
20.	Rebuild reverse index to normal index.
21.	Destroy index.
22.	Create type.
23.	Use type in table.
24.	Insert value for type in table.
25.	Display the table.
26.	Create array for name.
27.	Use array as data type in table.
28.	Insert value for array in table.
29.	Display table.

30.	Create user.
31.	Give grants to the user.
32.	Give grants to the user from own database.
33.	Give grant to a user from another user.
34.	Withdraw once given grant.
35.	Commit work done.
36.	Rollback to commit.
37.	Create savepoint.
38.	Rollback to the particular savepoint.